The “new ecology”

The holistic view of Earth involves studying the systems of organisms as a whole. As Odum explained, one organism, or any one group of organisms, cannot be understood without studying the ecosystem in which it lives. The holistic approach examines all the roles played by each member of an ecosystem, and how that system interacts with others. Climate, geology, water and mineral input, and human activity all affect—and are affected by—a multitude of living communities.

Odum was writing in the 1950s and ’60s, when there was a growing awareness of the environmental destruction wrought by humanity. The role of people was a crucial part of “systems ecology,” as he called his idea. Odum wanted humans to be sympathetic allies with the natural world—collaborators rather than manipulators—and his views of an all-embracing ecology did much to inspire the first Earth Day, which was celebrated in 1970.

The holistic concept of Odum’s “new ecology” deals with Earth as a whole, bringing together physics, chemistry, botany, zoology, geology, and meteorology. The fundamental assumptions of ecology are that the ecosystem is the basic unit of nature, that biological diversity increases the ability of ecosystems to survive, and that the whole is greater than the sum of its parts. Systems in the natural world—whether they are groups of cells in an animal’s body, the whole animal, or the ecosystem in which the animal lives—are able to self-regulate to provide stability.

“…ecology has been badly presented and has been broken into too many antagonistic subdivisions.”

Eugene Odum